

AI applications

Yaran Chen
Yaran.chen@xjtlu.edu.cn

AI: Face recognition



AI: Image recognition



License plate number recognition technology supports intelligent transportation applications



New AI Technology for Intelligent Marking (Source: CCTV-10 Science and Education Channel)

AI applications in life:

smart navigation system

Map navigation applications leverage artificial intelligence to achieve functions like route planning, real-time traffic information updates, and voice navigation prompts, helping people travel more conveniently

Smart home devices

Smart speakers, smart lamps, and smart home appliances can be remotely controlled and have automated scene Settings through mobile phone apps or voice commands, bringing convenience to people's lives.

AI

intelligent recommendation system

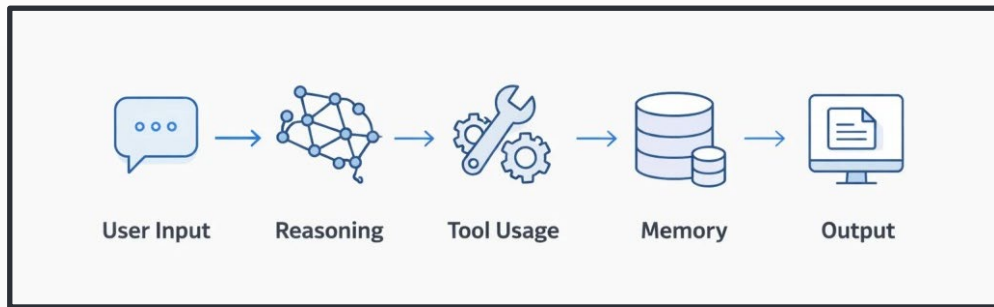
E-commerce platforms and music platforms will, based on users' browsing history, purchase records, collection preferences and other data, use artificial intelligence algorithms to recommend personalized products, videos, music

Intelligent security monitoring

Smart cameras installed in residential areas, shopping malls, homes and other places can perform face recognition, behavior analysis and other functions through artificial intelligence technology. For instance, when abnormal behaviors

AI Systems

A Typical Workflow for AI Systems



- 1. Formulate Problem:** Define the problem, establish goals, and translate business/research context into a well-defined AI task.
- 2. Collect Data:** Gather relevant, high-quality data, including sourcing, cleaning, and preprocessing.
- 3. Feature Extraction:** Identify and select informative attributes from raw data for machine learning.
- 4. Machine Learning:** Apply algorithms to train a model, tune parameters, and validate performance.
- 5. System Deployed:** Integrate the model into production, monitor, and maintain it for real-time use.

Hierarchy of AI Intelligence

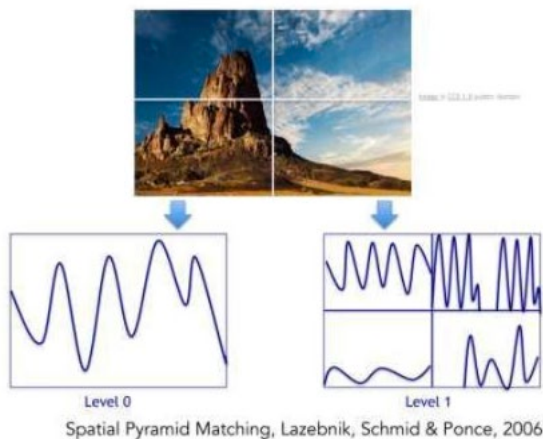
- **Computation Intelligence:** Maps Features to a unified Label. Key applications: Classification, Outlier Detection.
- **Perception Intelligence:** Maps Features to multiple, non-unified Labels. E.g., Emotion Detection.
- **Creative Intelligence:** Generates novel content beyond pattern matching, augmenting human creativity.
- **Universal Intelligence:** Hypothetical ability to learn any human intellectual task (AGI goal).



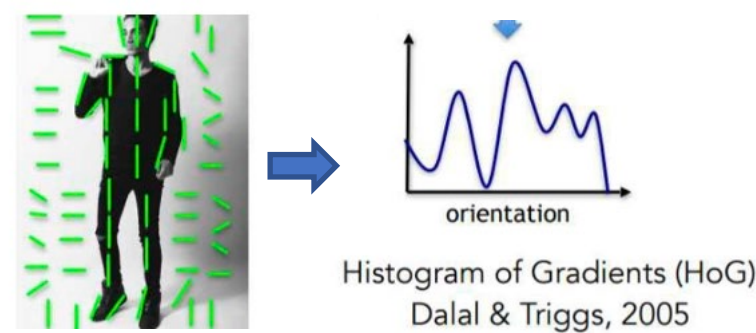
Feature Extraction



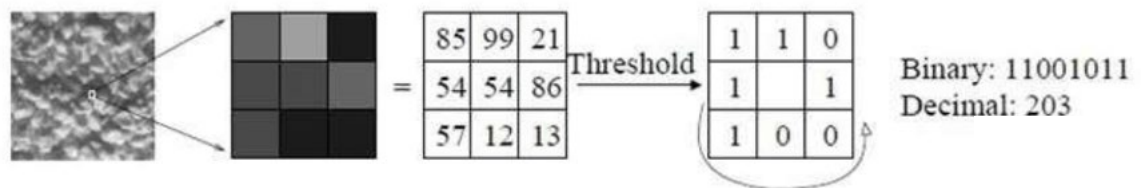
Haar Features



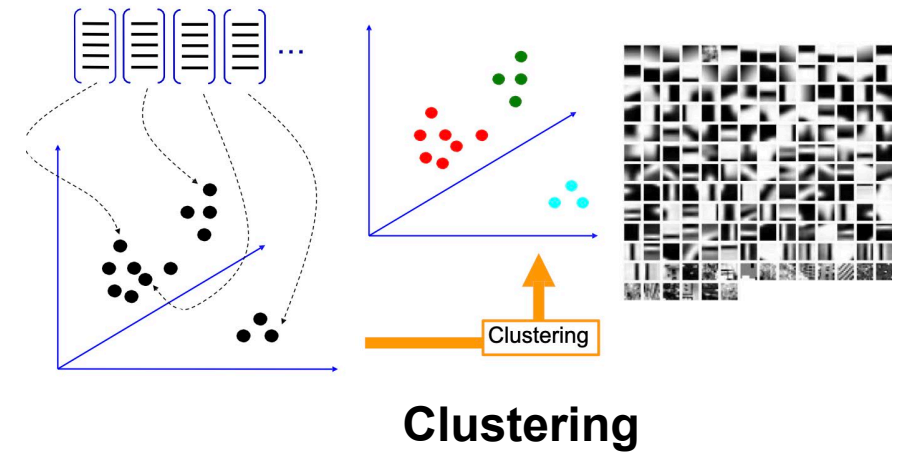
Spatial Pyramid Matching



Histogram of Gradients (HoG)



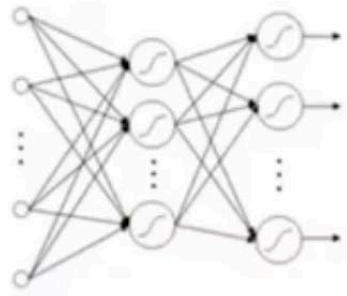
Local Binary Patterns (LBP)



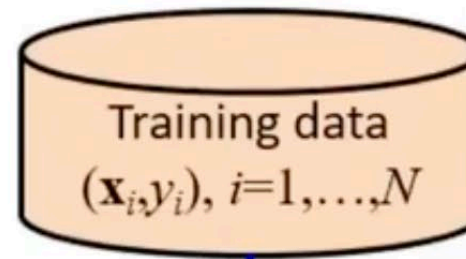
Clustering

Model Training

Prepare training data: Including feature representation and class labels



Classifier structure
 $g_i(\mathbf{x}, \theta_i), i=1, \dots, M$



Parameter estimation
min. loss or max. likelihood

Parameter values Θ

Choose classifier type and determine hypothesis parameters: represented as models for each class or discriminant functions

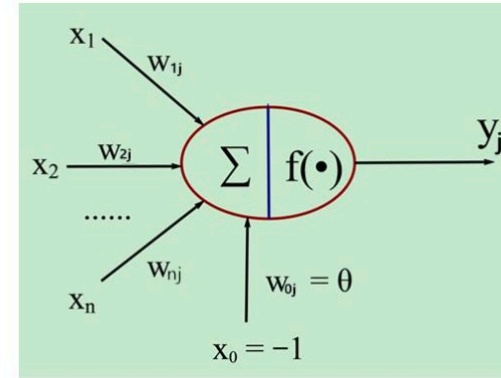
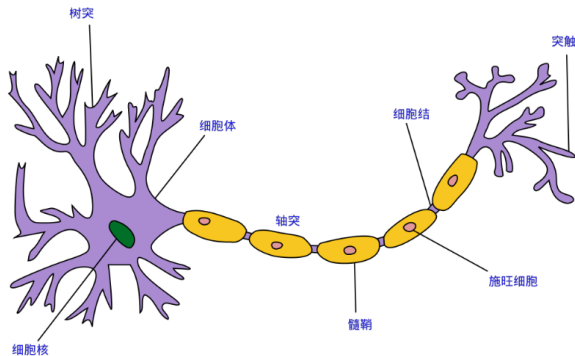
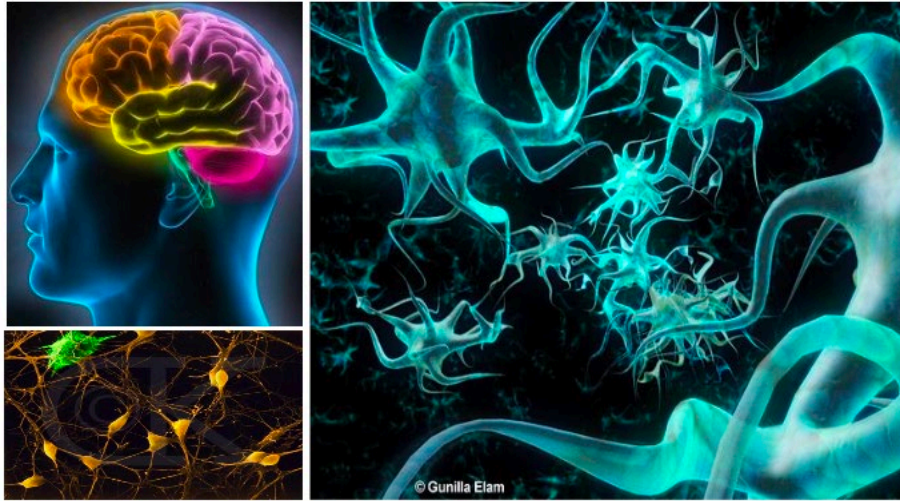
Loss function
 $l(\mathbf{x}, y, \Theta)$

$$\min E = \frac{1}{N} \sum_{i=1}^N l(\mathbf{x}_i, y_i, \Theta)$$

Minimize loss or maximize likelihood

Neural Networks

The human brain contains many (approximately 80 billion) neurons.
Each neuron consists of two parts: the cell body and dendrites.



$$net_j(t) = \sum_{i=1}^n \omega_{ij} \chi_i(t) - \theta_j$$

If the threshold is considered as an input x_0 with weight w_{0j} :

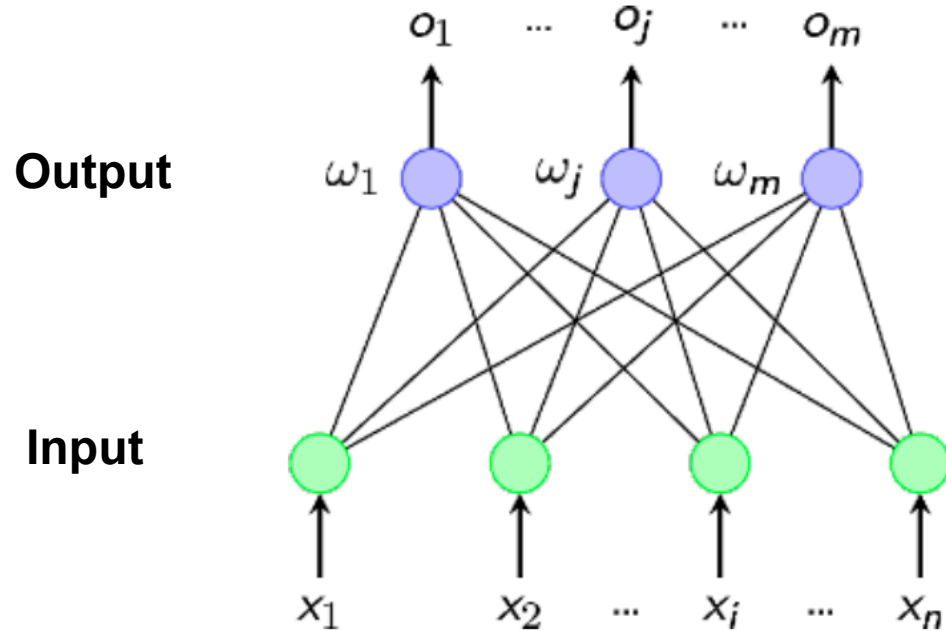
$$net_j(t) = \sum_{i=0}^n \omega_{ij} \chi_i(t) = W^T X$$

$$y_j = f(net_j) = \begin{cases} 1, & net_j \geq 0 \\ 0, & net_j < 0 \end{cases}$$

y_j represents the output of neuron j , and f represents the activation function.



Neural Networks



$$\sigma_j = \text{sgn}(W_j^T X) = \begin{cases} 1, & W_j^T X > 0 \\ 0, & W_j^T X = 0 \\ -1, & W_j^T X < 0 \end{cases}$$

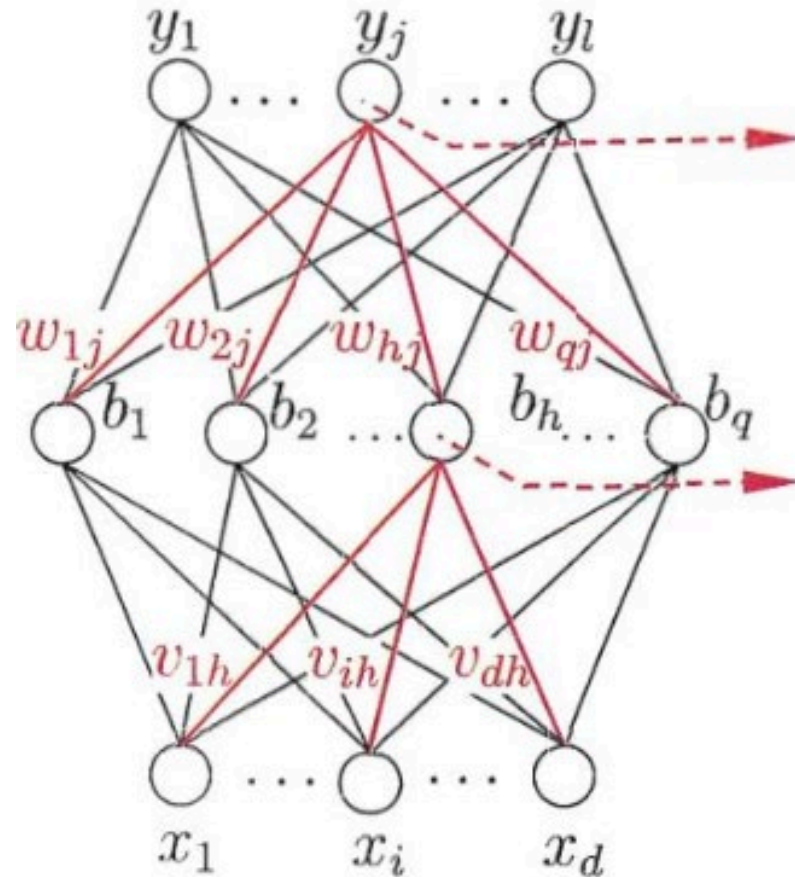
Single-layer Neural Network

Neural Networks

Output Layer

Hidden Layer

Input Layer



Input to the j -th output neuron:

$$\beta_j = \sum_{h=1}^q w_{hj} b_h$$

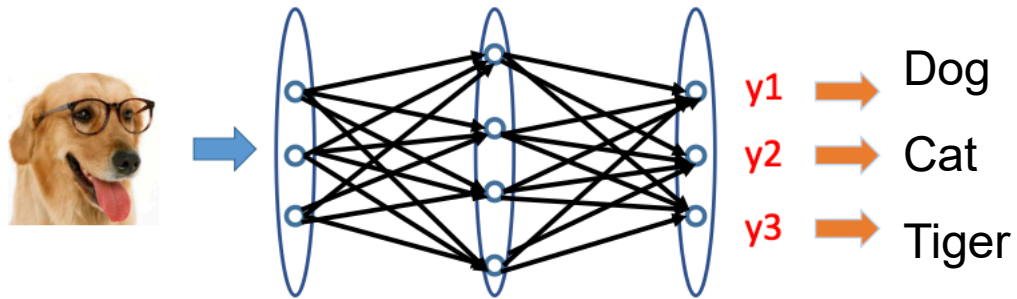
Input to the h -th hidden neuron:

$$\alpha_h = \sum_{i=1}^d v_{ih} x_i$$

Multi-layer Neural Network

Neural Networks

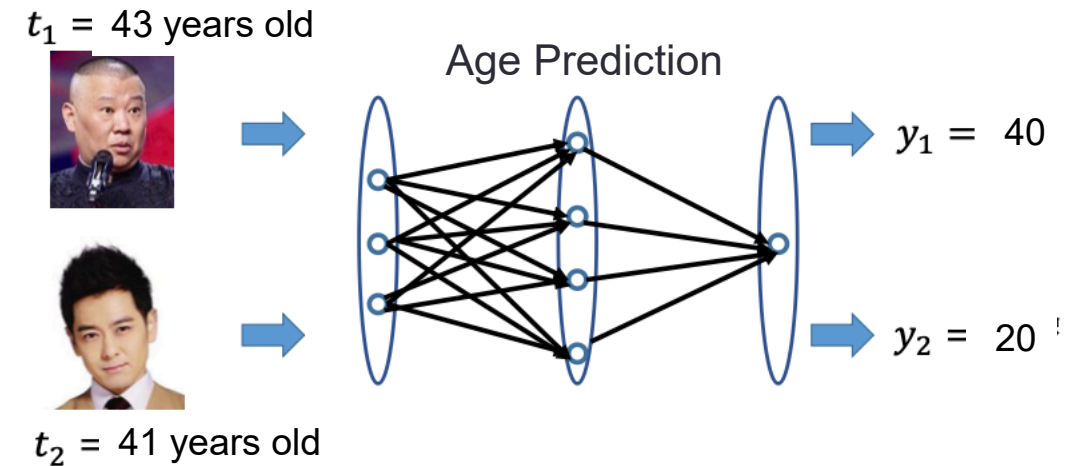
Classifications



softmax
$$p_1 = \frac{\exp(y_1)}{\exp(y_1) + \exp(y_2) + \exp(y_3)}$$

Loss function
$$loss = -\log p_1$$

Predictions



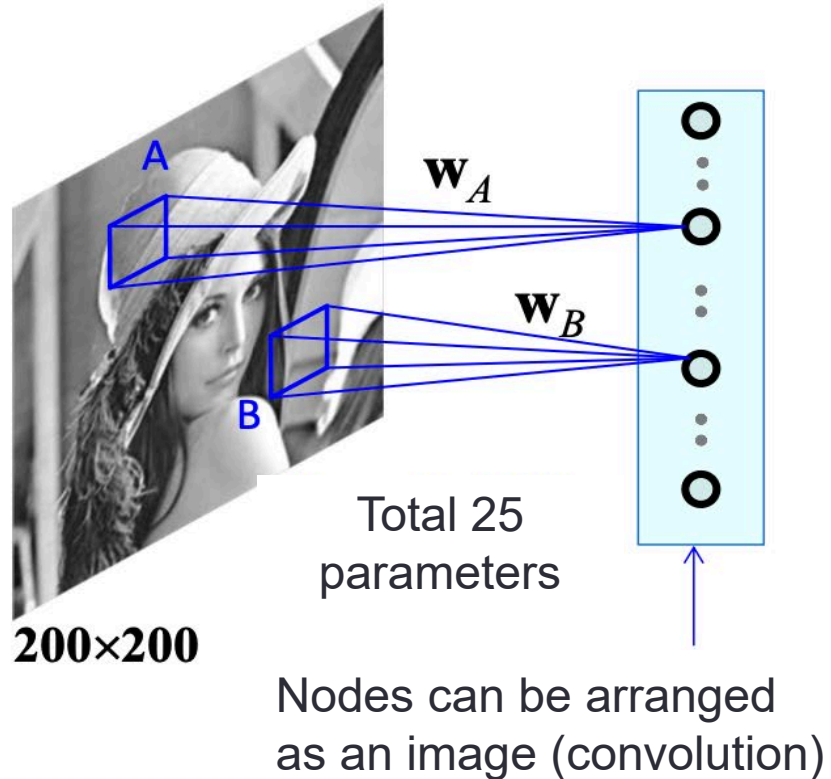
Loss function
$$loss = (y_1 - t_1)^2 + (y_2 - t_2)^2 + \dots$$



Convolutional Neural Networks

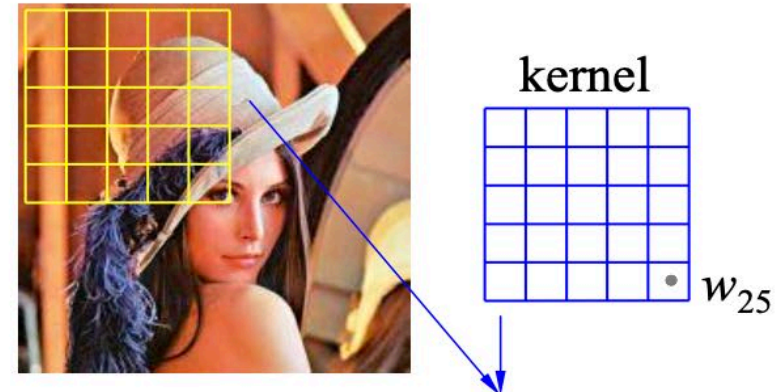
Convolution Operation Using Local Structures in Images

Consider a 5x5 window



Convolution Operation

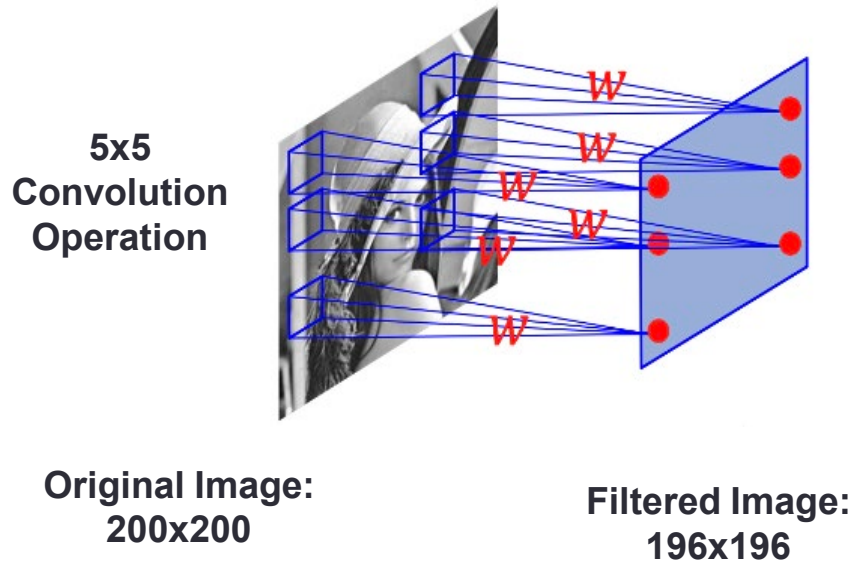
Image * filter



Weighted Sum

$Wx+b$

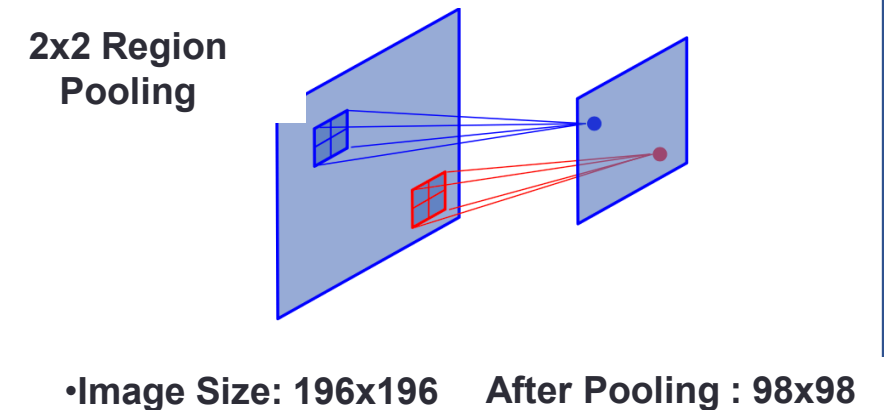
Convolutional Neural Networks



Using Local Connections + Weight Sharing:

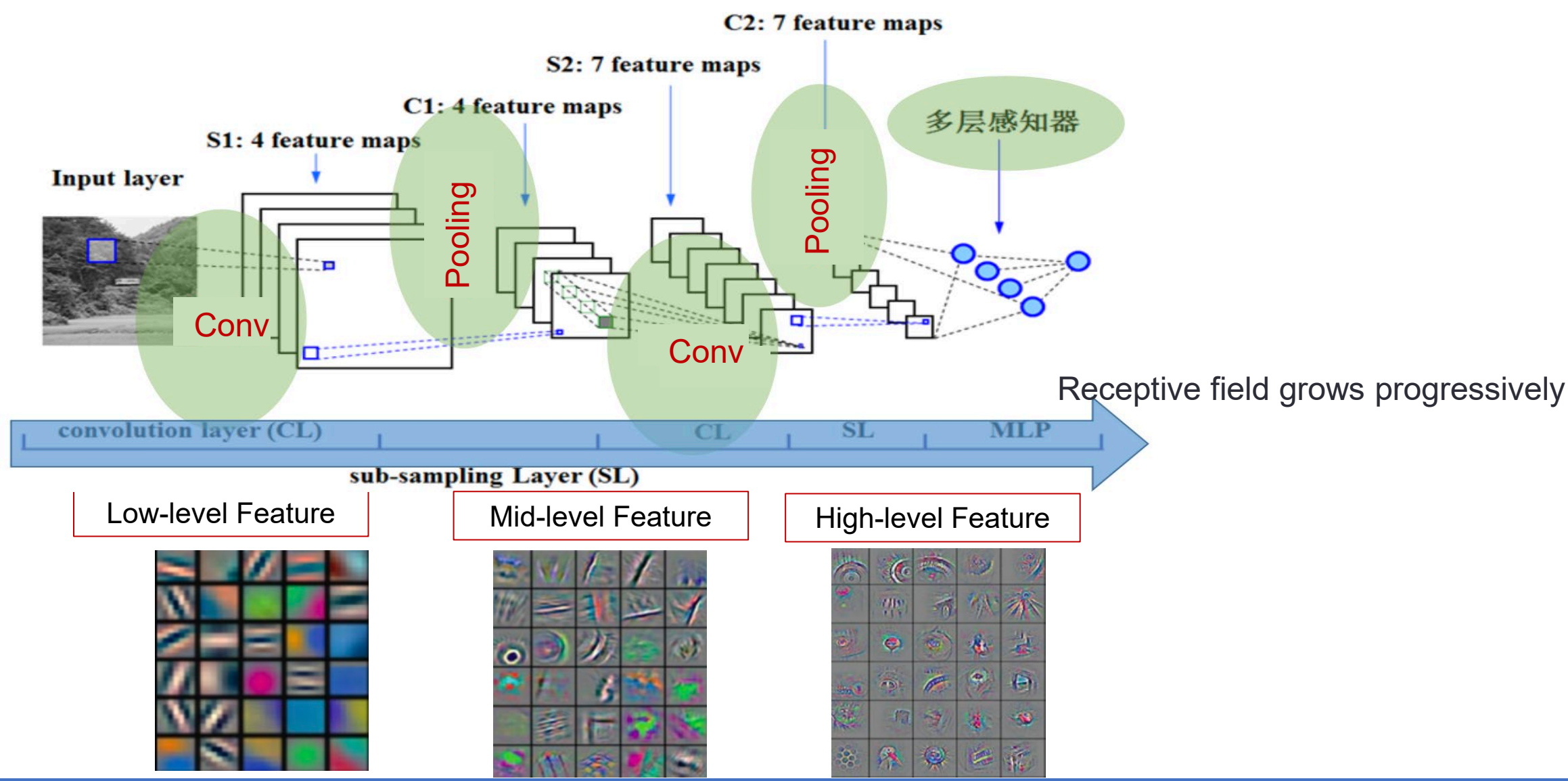
- Total 25 Parameters!
- Weight Sharing, Translation Invariant

Pooling: Enhancing Network Robustness, Invariance to Deformations and Translations, Increasing Receptive Field



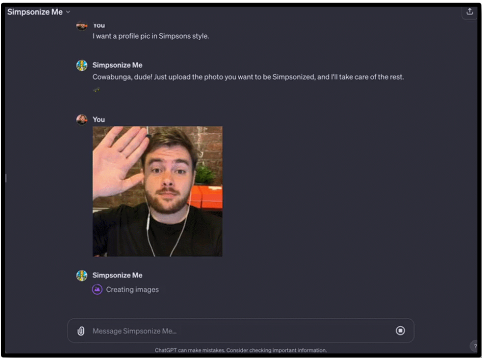
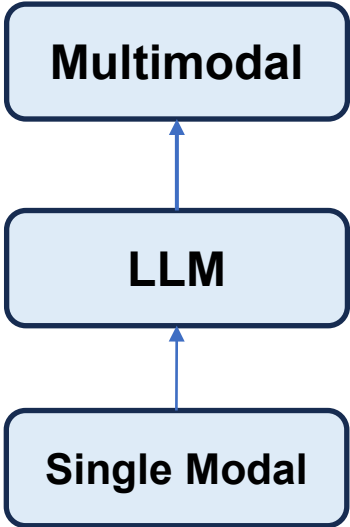
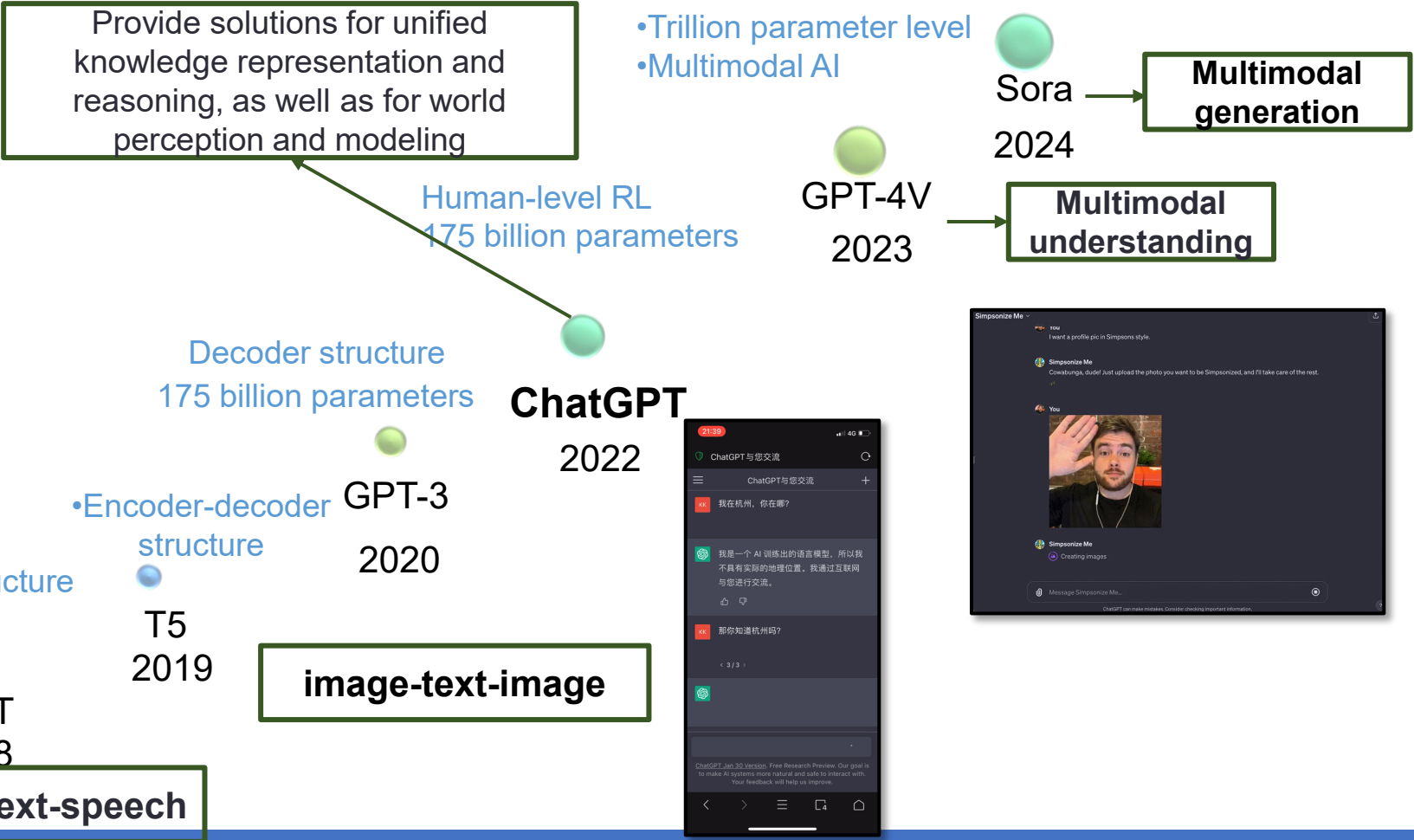
Average pooling: Take the Average
Max pooling: Take the Maximum

Convolutional Neural Networks



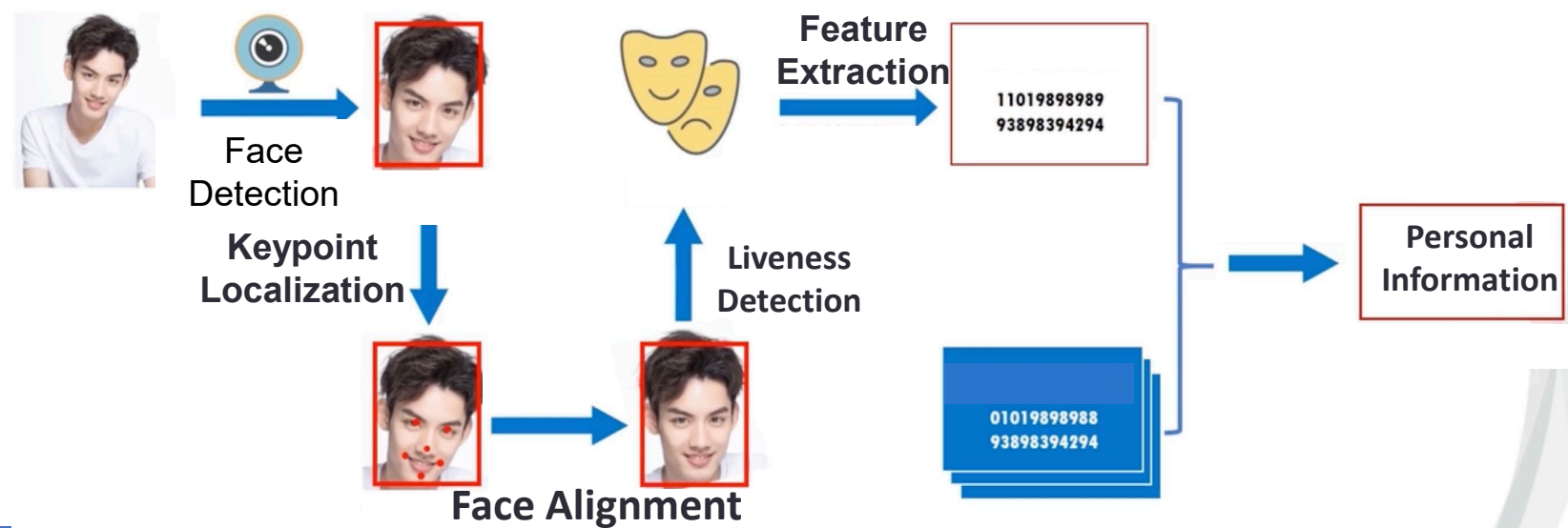
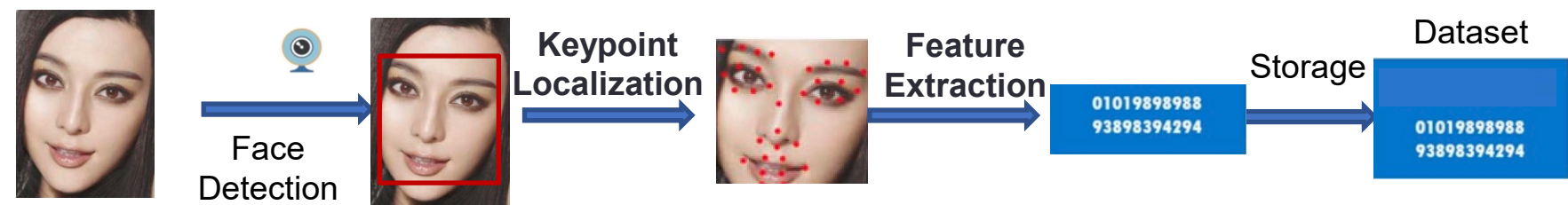
Transformer

Supporting Parallel Processing Technology
Attention Module models long-term dependencies
Provides a unified architecture for vision and language processing



Application: Face Recognition

Extract Key Points Features
Image-to-Face Recognition Matching with a Face Database

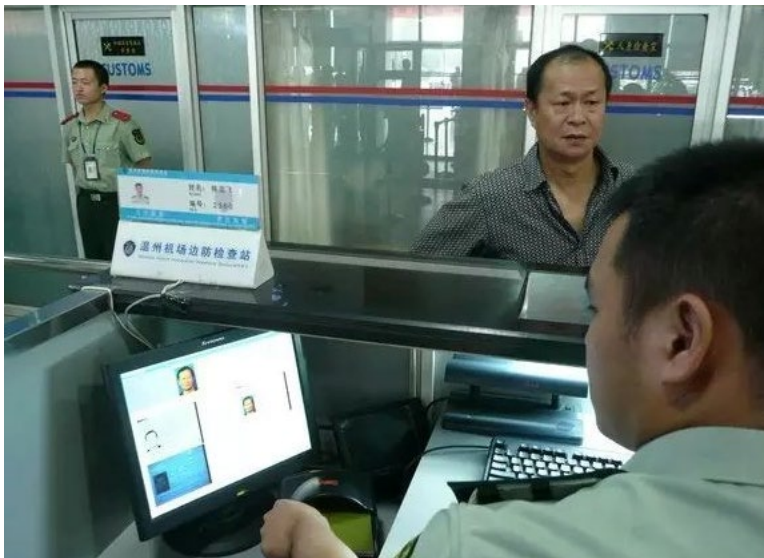


Application: Face Recognition

- **Face Detection:** Locating the position of faces in images.
Haar feature-based classifiers, DPM, and DNNs like MTCNN.
- **Keypoint Localization:** Determining key facial features such as eyes, nose, and mouth.
DNNs, such as Dlib's facial landmark detector or other CNN-based methods.
- **Face Alignment:** Aligning different faces to the same pose and position for subsequent processing.
keypoint localization, use geometric to align faces.
- **Liveness Detection:** Determining whether the input face is from a real person rather than a photo or video.
DNNs combined with multiple sensors for multimodal analysis, or detect actions.
- **Face Feature Extraction:** Converting face images into fixed-length feature vectors.
DNNs such as FaceNet and ArcFace to extract feature vectors from faces.
- **Feature Value Comparison:** Finding the most similar feature vectors to identify corresponding faces.
Distance metrics to compare the extracted feature vectors with those in the database.
- **Personal Information:** Completing the final face recognition task.
Retrieve personal information from the database.

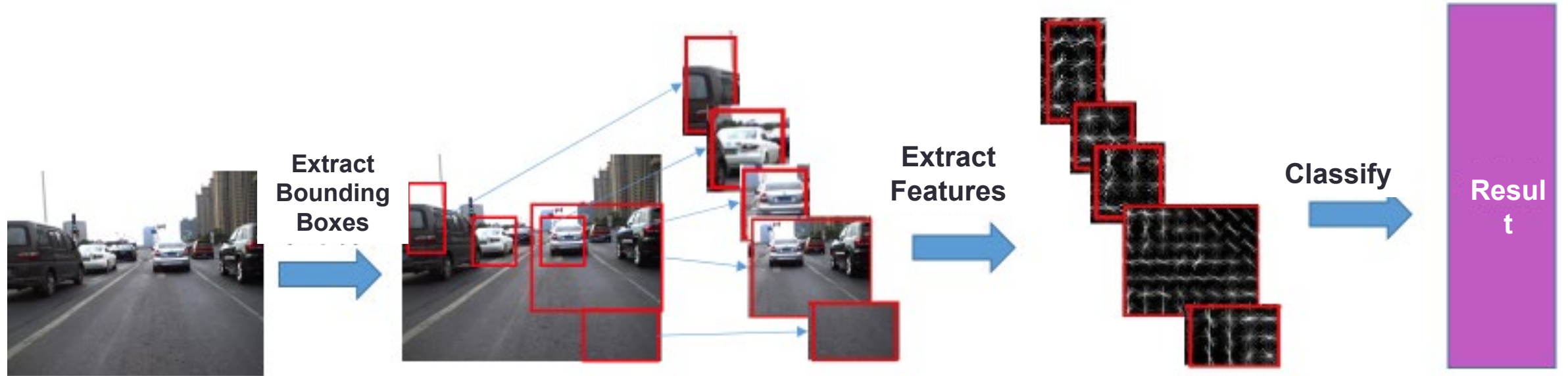


Application: Face Recognition



Application: Object Detection

Object Detection Methods Based on Machine Learning



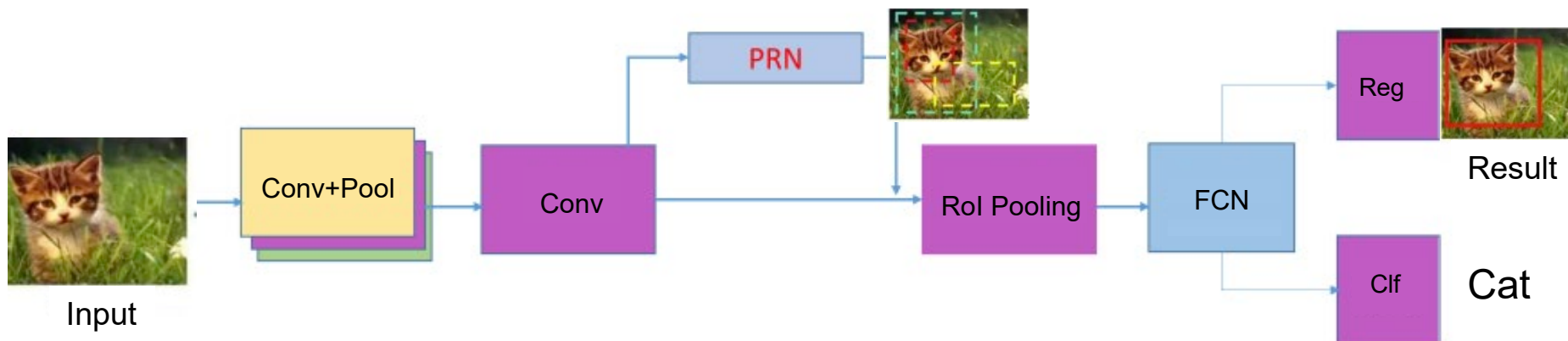
- Fast speed, simple
- Manually designed features, poor performance in complex scenes, dim lighting, or occlusion

HOG
LBP
Haar-Like,
SIFT
.....

SVM
Decision Trees
Random Forests
Boosting
.....

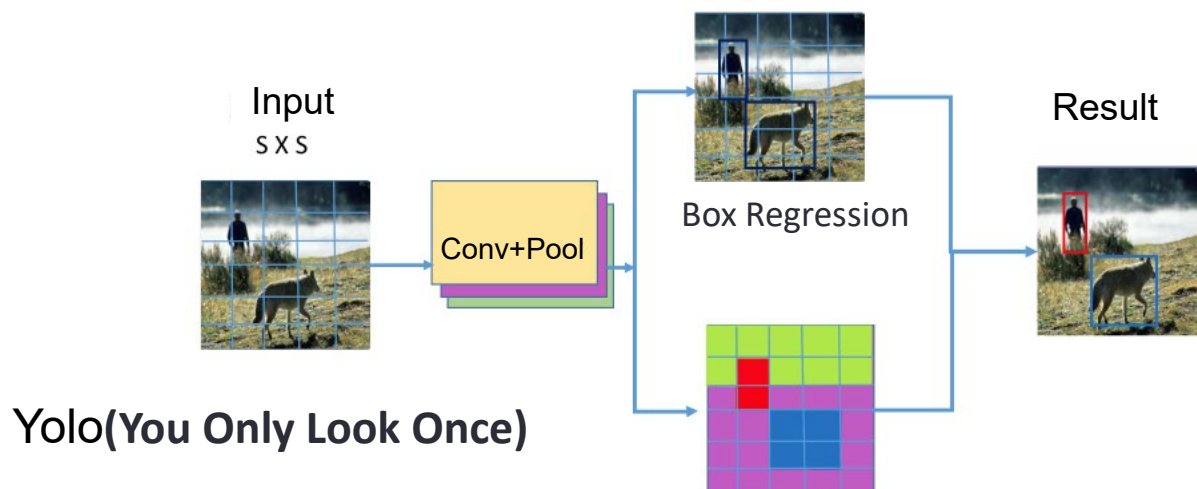
Application: Object Detection

Object Detection Methods Based on Machine Learning



Two-Stage Detection:

- **Fast RCNN**
- **Faster RCNN**
- Extract Regions of Interest (RoIs)
- Detect and Classify RoIs

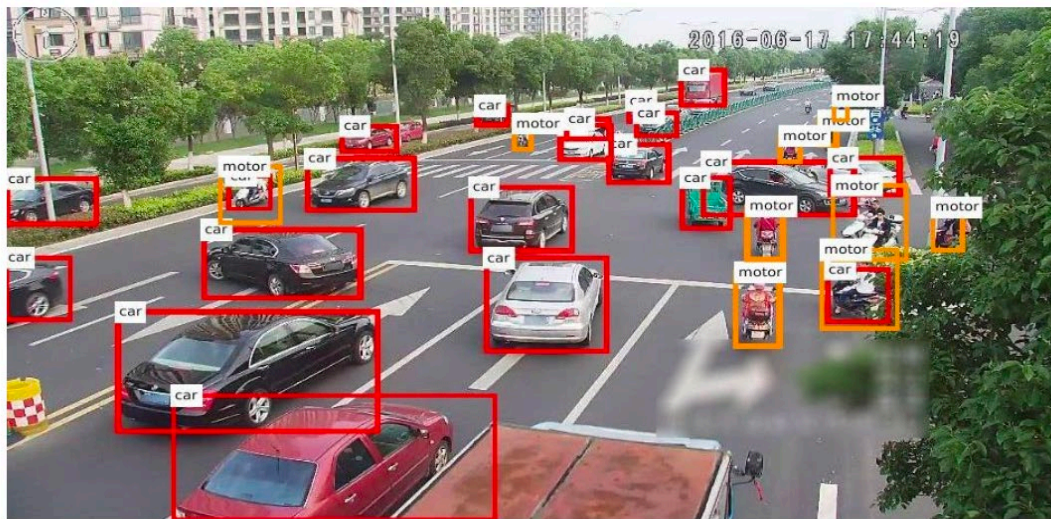


One-Stage Detection:

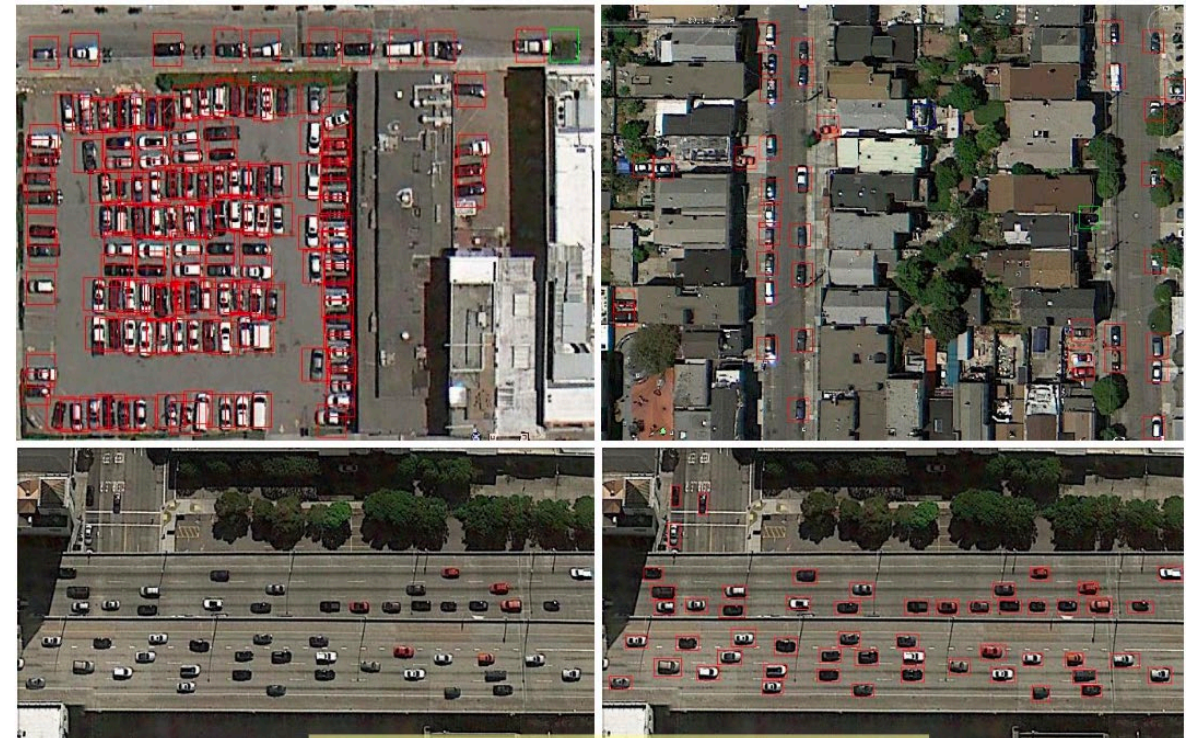
End-to-end output of detection box regression and classification results

Application: Object Detection

Vehicle Object Detection

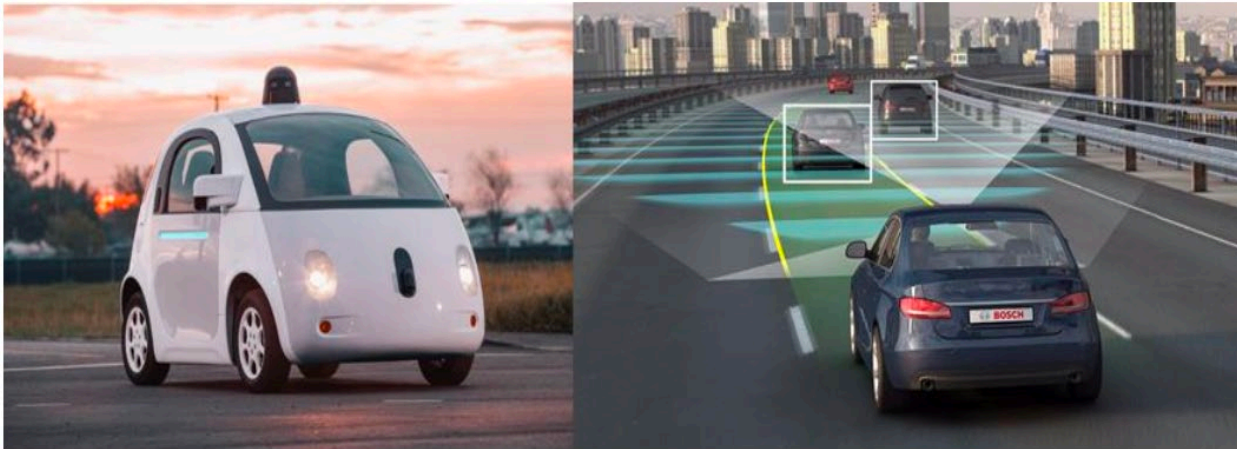


Chang, ITSM 2017



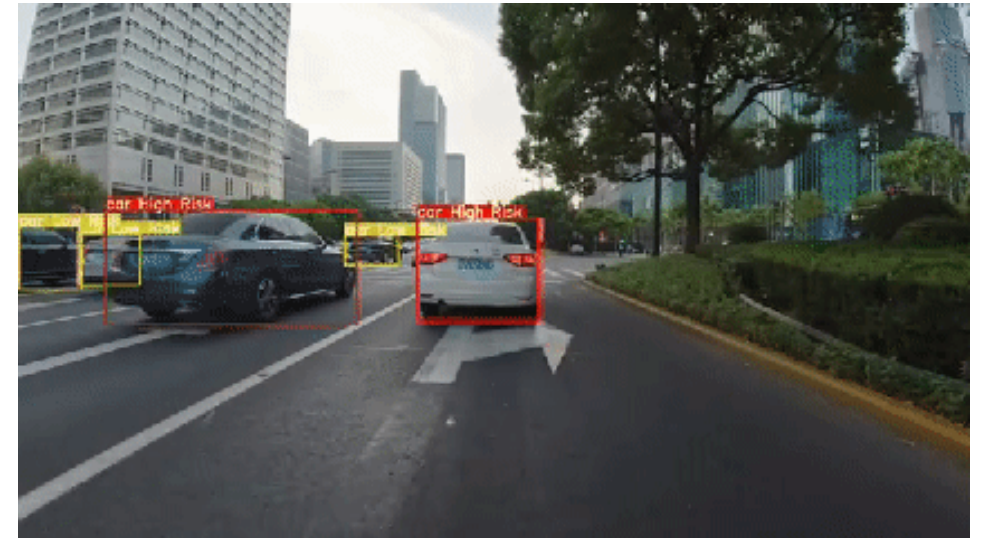
Application: Object Detection

Autonomous Driving



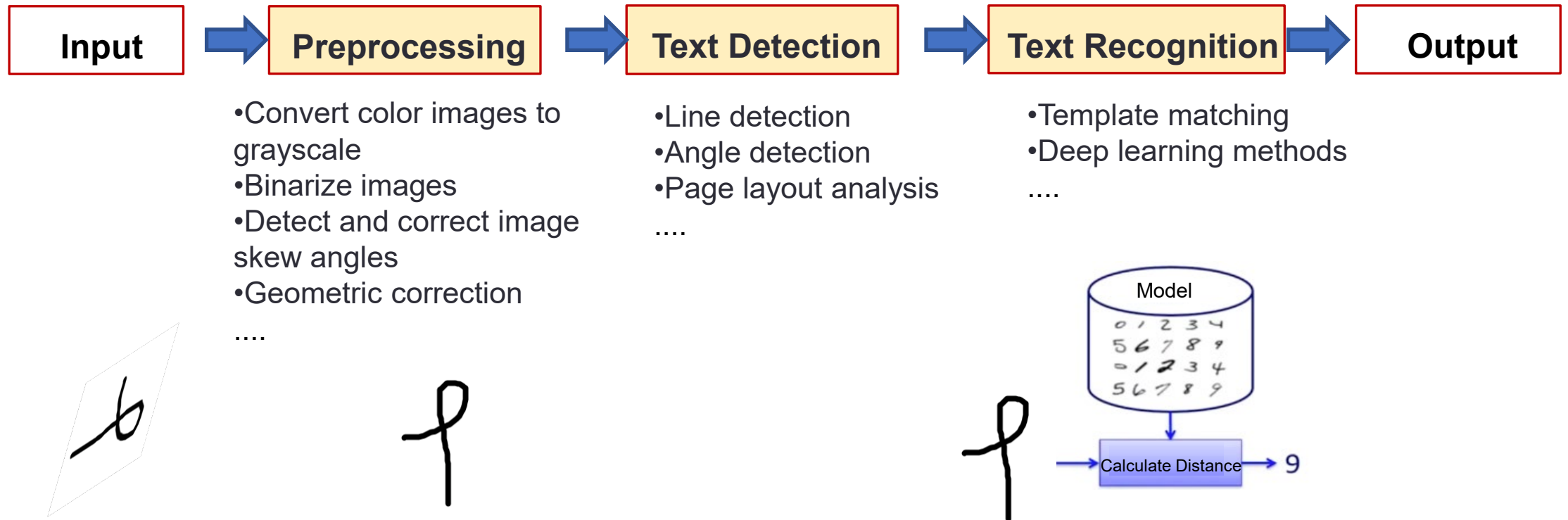
Mariusz Bojarski, Davide Del Testa, Daniel Dworakowski, et al.. End to End Learning for Self-Driving Cars, arXiv:1604.07316, 2016

<http://news.bitauto.com/hao/wenzhang/720984>



Application: Image-Text Detection

Optical Character Recognition, OCR:



Application: Image-Text Recognition

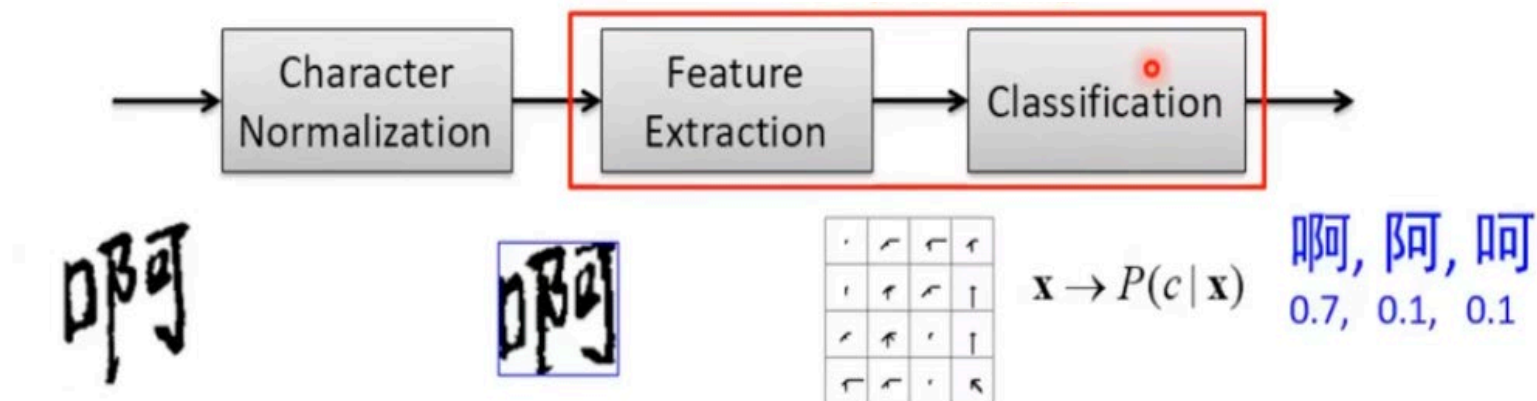
Optical Character Recognition, OCR:

Text Recognition

Feature Extraction: HOG/ Deep Features

Classifier: Provide candidate categories.

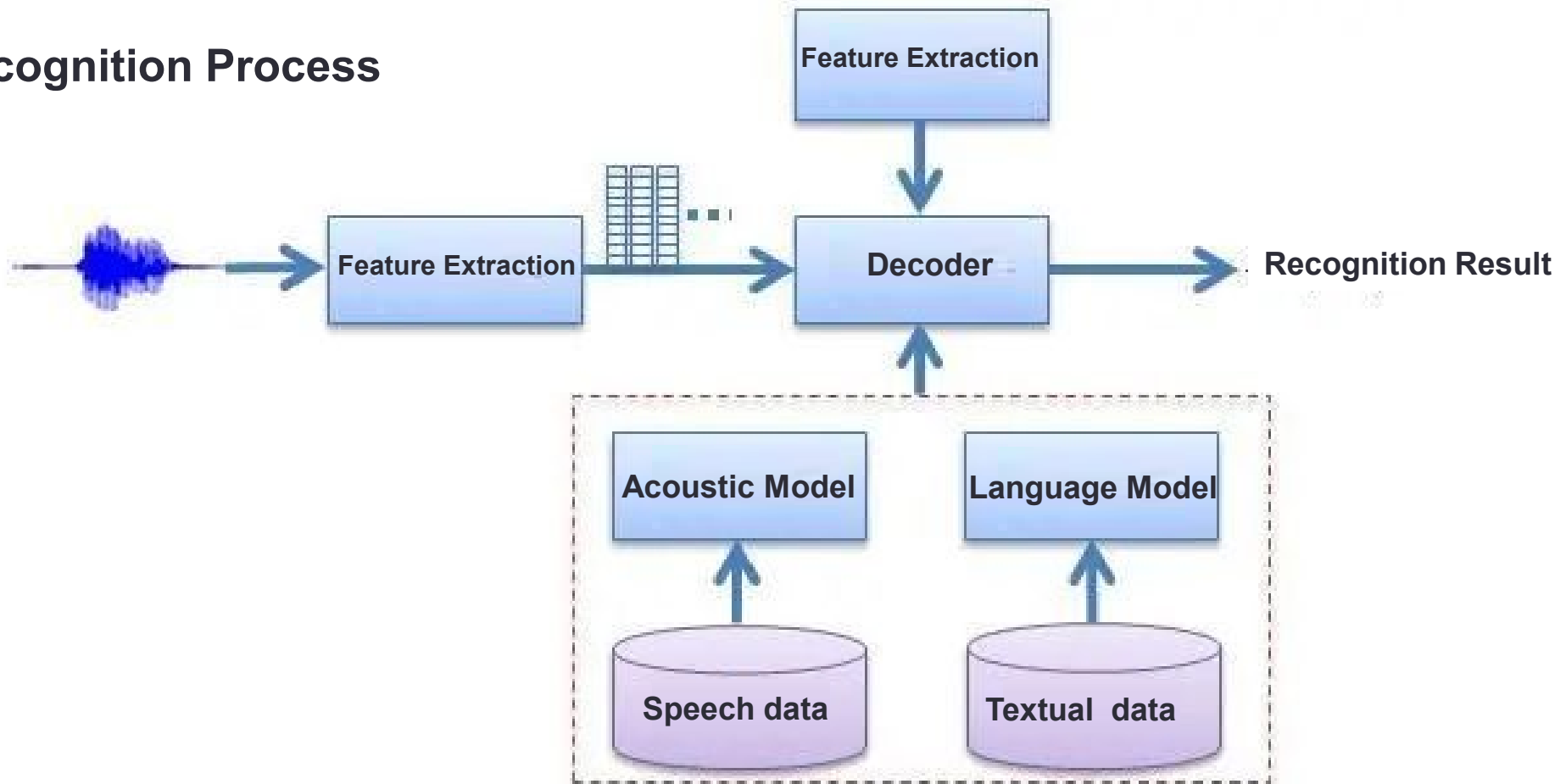
- Statistical Classifiers, Support Vector Machines (SVM), NN, etc.
- **DNN:** Joint learning of features and classifiers.





Application: Speech Recognition

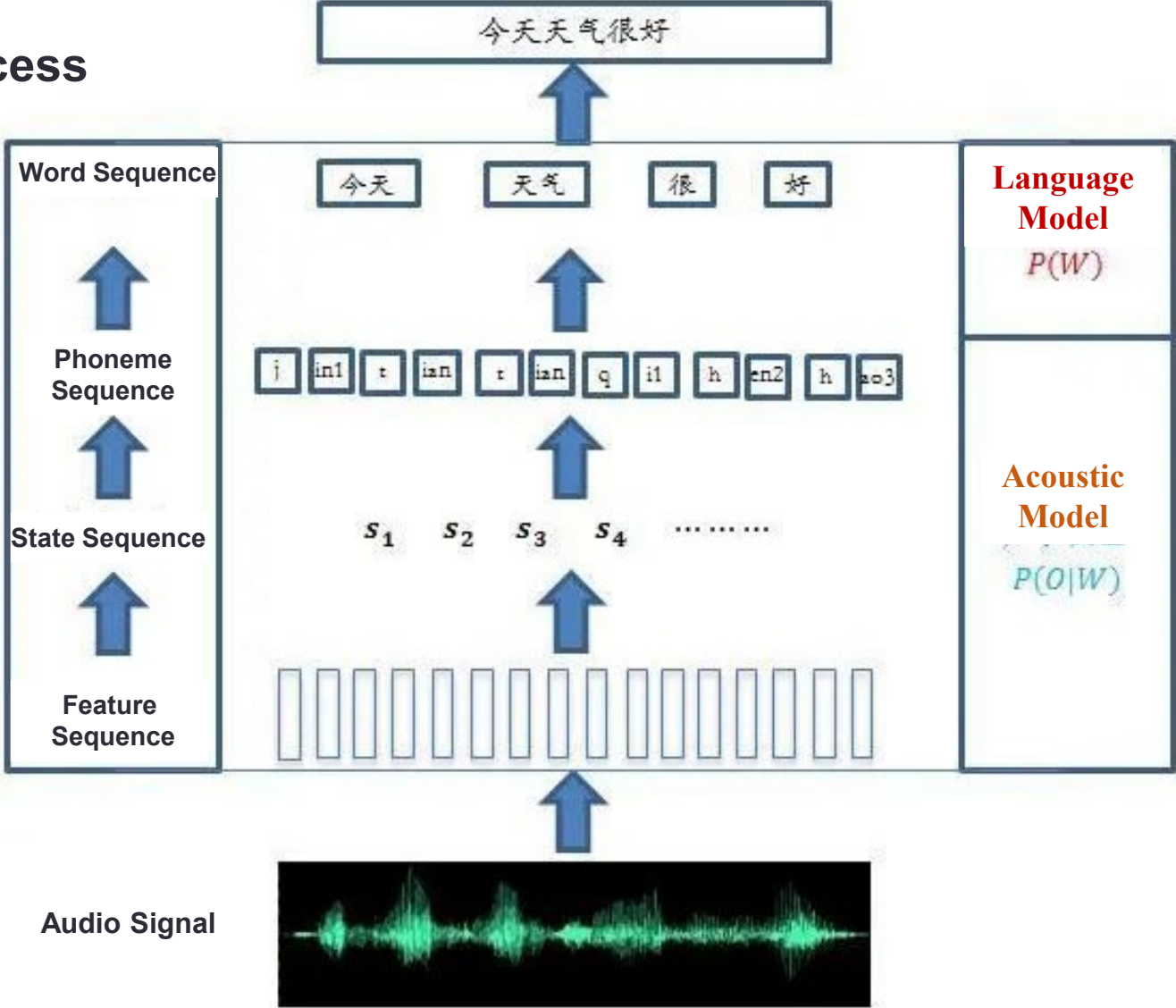
Speech Recognition Process



- **Acoustic Model:** Trained on large amounts of speech data (covering various accents, ages, genders, speaking rates, etc.) using GMM-HMM/DNN-HMM
- **Language Model:** Trained on various textual corpora.

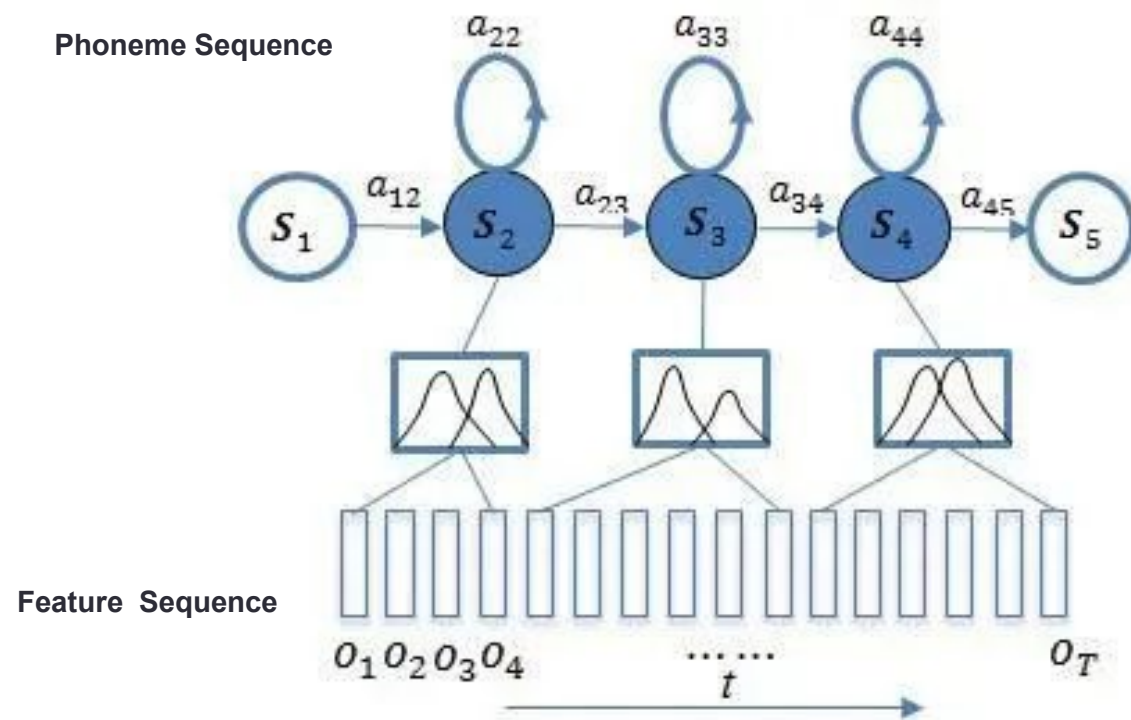
Application: Speech Recognition

Speech Recognition Process

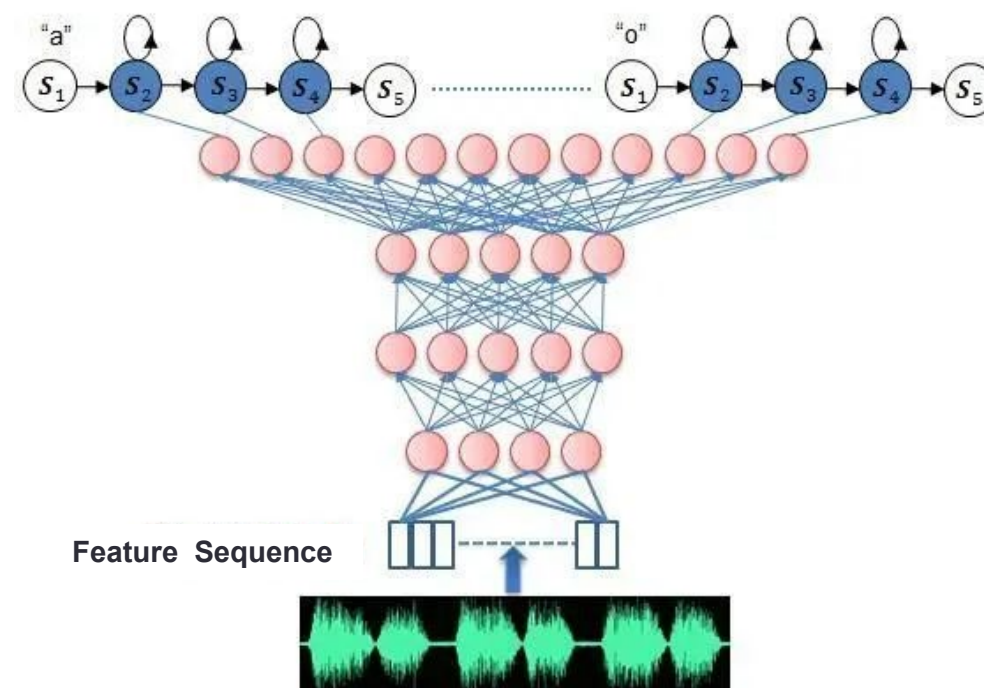


Application: Speech Recognition

Acoustic Model: GMM-HMM, DNN-HMM

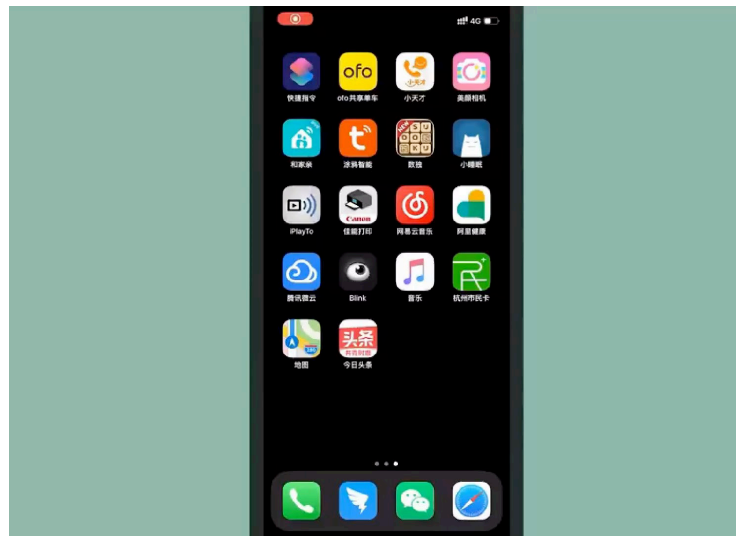


GMM-HMM



DNN-HMM

Application: Speech Recognition



	官方定位	人物设定	当前特色技能
 Siri	iOS 设备中集成的语音控制程序,令设备化身智能机器人	喜欢有层次的绿色的苹果小公主	基于手机设备中的私人信息,展开 proactive 主动提醒
 灵犀	最懂你声音的智能语音助手	未来平行世界人工智能的宠萌幼年体,射手座	预言术,主动提醒;读心术,贴心聊天
 小娜	微软发布的第一款个人智能助理	来自 500 年后,士官长女朋友	可以装在 Win10 电脑上提供建议
 小冰	陪聊机器人	淘气邻家女孩,处女座	识图测颜值
 度秘	让 Siri 哭晕在厕所的秘书	性别女,可爱机器人	提供 O2O 服务

Application: Text-to-Speech

Text-to-Speech Process

Text Analysis

Extract pronunciation and prosody

- Text normalization
- Phoneme conversion
- Prosody prediction

Acoustic Model

Generate acoustic features

- Tacotron 1/2
- Deep Voice 1/2/3
- Transformer TTS
- FastSpeech/FastSpeech 2
- Glow-TTS
- BVAE-TTS

Vocoder

Synthesize speech signals

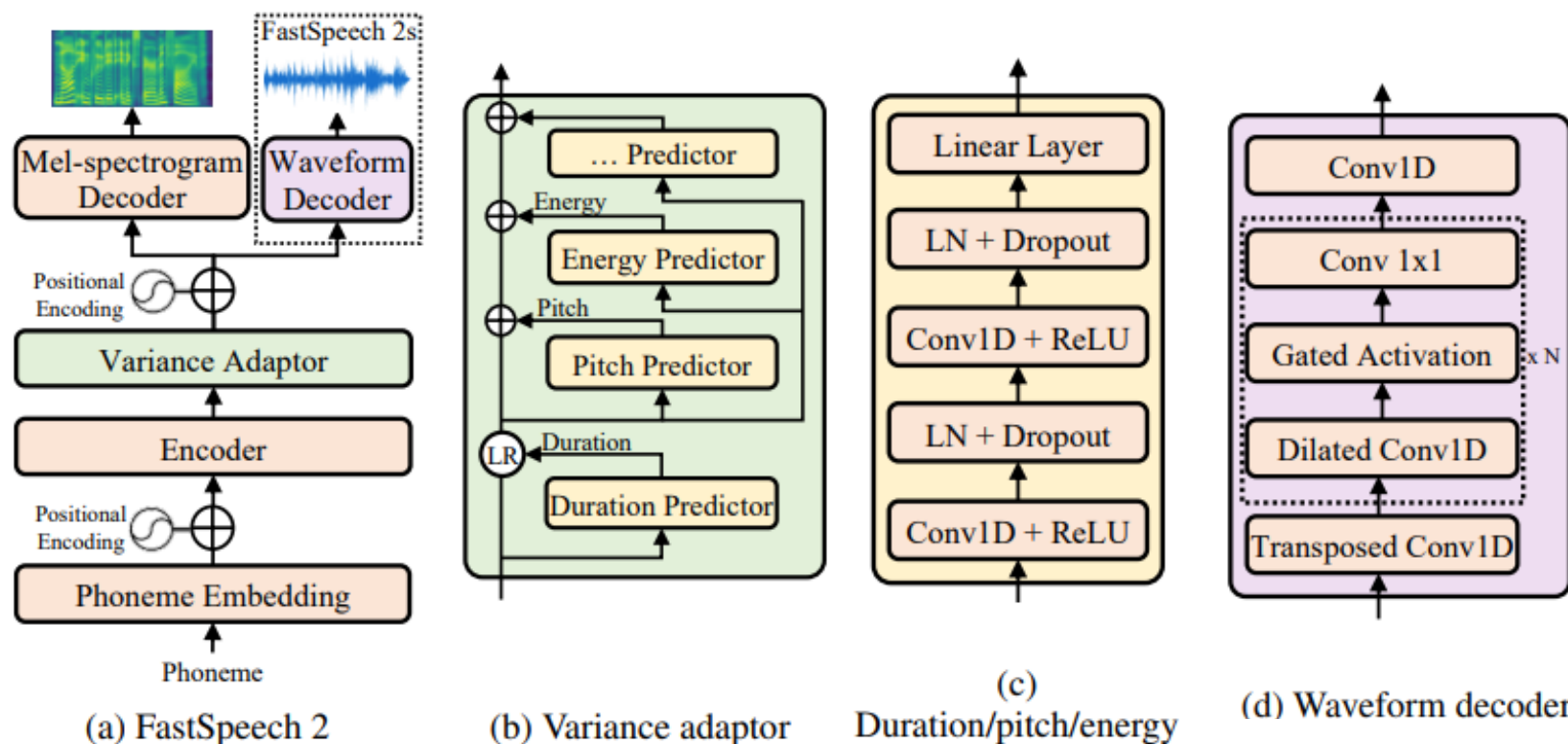
- Griffin-Lim
- WORLD/STRAIGHT
- WaveNet/Parallel WaveNet
- WaveGlow/FloWaveNet
- WaveRNN/LPCNet



Both text analysis and vocoder have some general solutions, while the main focus of works is acoustic model.

Application: Text-to-Speech

Transformer-based Acoustic Model



FastSpeech2, ICASSP 2021

Application: Text-to-Speech



Human-Machine Interaction
Personalization



AIGC Content Creation
Emotion Accent Dialect Language



Metaverse
Virtual Humans/Avatars



Generation and Verification

Machine Learning vs. Deep learning



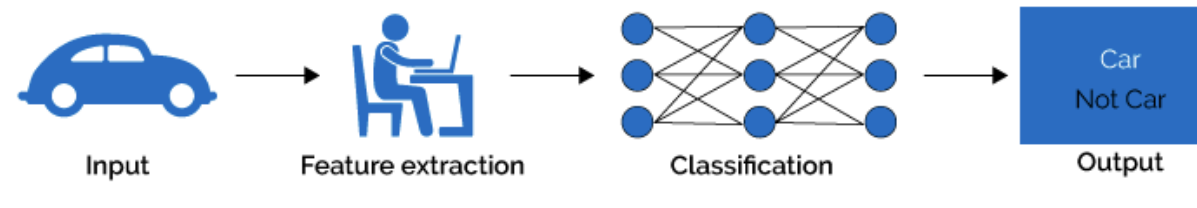
Machine Learning (ML)

Requires human experts to define features (e.g., "has wheels"). The computer then classifies based on this predefined checklist. This is called manual feature engineering.

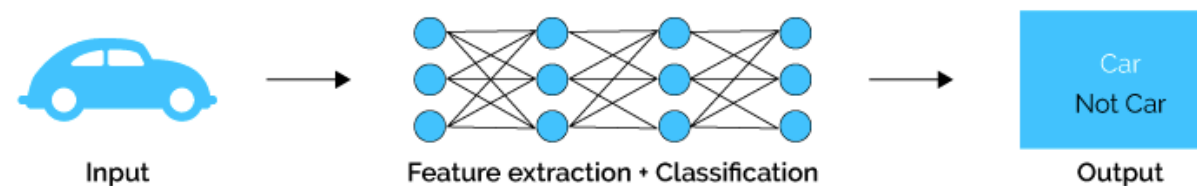
Deep Learning (DL)

Automatically learns features from raw data using a Neural Network. It excels at complex tasks like image and speech recognition, removing the need for manual feature engineering.

Machine Learning



Deep Learning



The two phases of AI

First-Generation AI (Symbolism)

Core Principle: Simulates logical reasoning by manipulating human-readable symbols.

Key Driver: **Knowledge** (Expert Systems, Rules)

Key Tech: Knowledge Graphs, Inference Engines



Example: IBM DeepBlue

Defeated Chess World
Champion Kasparov (1997)

Second-Generation AI (Connectionism)

Core Principle: Intelligence emerges from the interaction of simple, interconnected units.

Key Driver: **Data** (Statistical Learning, Big Data)

Key Tech: Neural Networks, Large Language Models (LLMs)



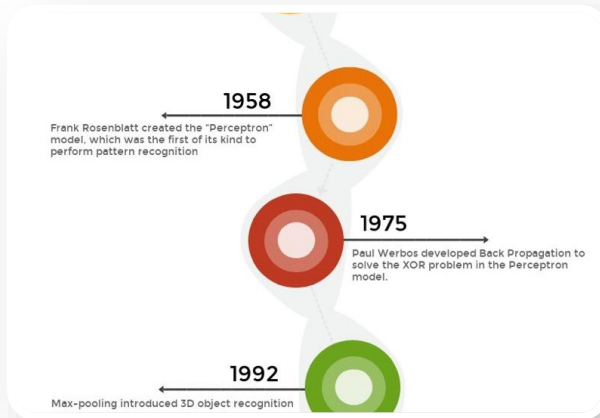
Example: AlphaGo

Defeated Go World Champion
Lee Sedol (2016)



The Core of AI – Neural Network Algorithms

Evolution Timeline



1950s: Machine Learning emerges.

1990s: Neural Networks (CNN, RNN, GAN) flourish.

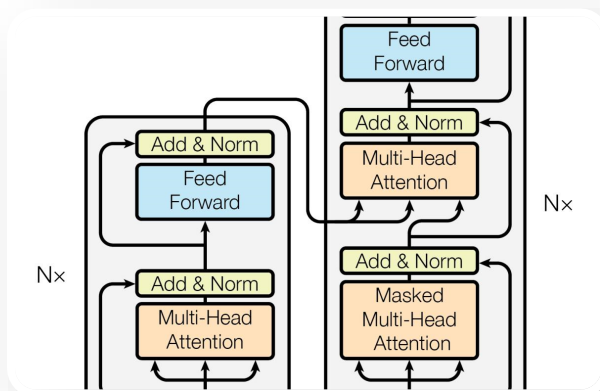
2017: Transformer architecture proposed.

2020: Rise of large-scale pre-trained models.

2022: ChatGPT released to the public.

2024: Emergence of Reasoning Large Models.

Key Models and Technologies



Machine Learning

Foundation of modern AI systems



Neural Networks

CNNs, RNNs, GANs



Transformer

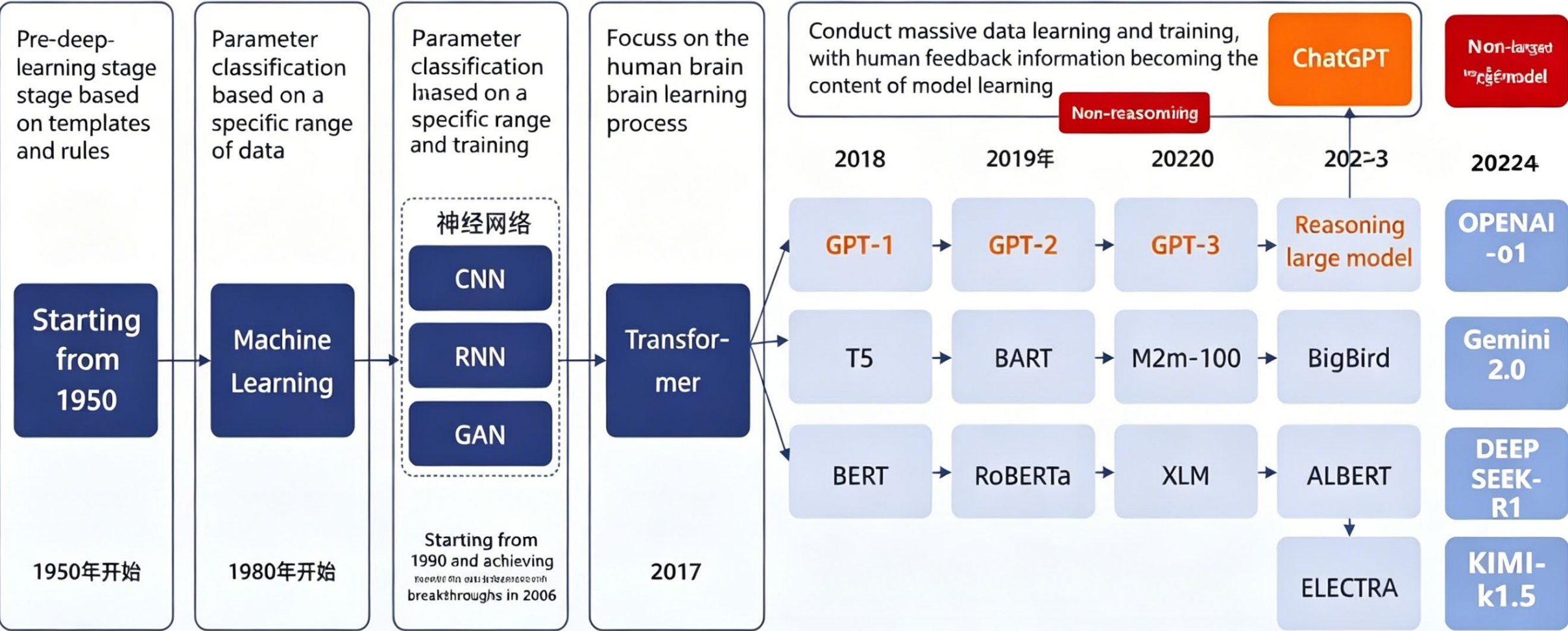
"Attention Is All You Need"



LLMs

GPT Series & BERT Series

The Core of Artificial Intelligence — Neural Network Algorithms



Deep neural network algorithms, the environment, paradigms and rules of large model training, engineering integration, business models, etc., are the result to comprehensive effects.

The Essence of Neural Networks: It's All About "Guessing"

$$Y = aX + b$$



$$Y = a_1X + a_2X^2 + \dots$$

Given $X=1, Y=2$; $X=2, Y=3$

Method: Use algebra.

Process: With two known points (x_1, y_1) and (x_2, y_2) , set up two equations and solve for the unknown variables 'a' and 'b'.

Result: An exact, analytical solution. This is the traditional method taught in schools and math classes.

Method: Guess and Check (Iterative Refinement).

Process: 1. Start with random initial values for 'a' and 'b'.

2. Calculate the error (the difference between the guess and the real value).

3. Use the error to refine the next guess.

4. Repeat this cycle thousands of times.

Result: An approximate, but often very good solution. This is how machine learning algorithms often work.

Machine Learning



Deep Learning

RNN: 循环神经网络, 循环串行猜 (用前一个字猜后一个字)。但长文本算力限制, 无法逼近



DCNN: A Radical Shift Every element in the input is connected to every other element, regardless of their position in the sequence.

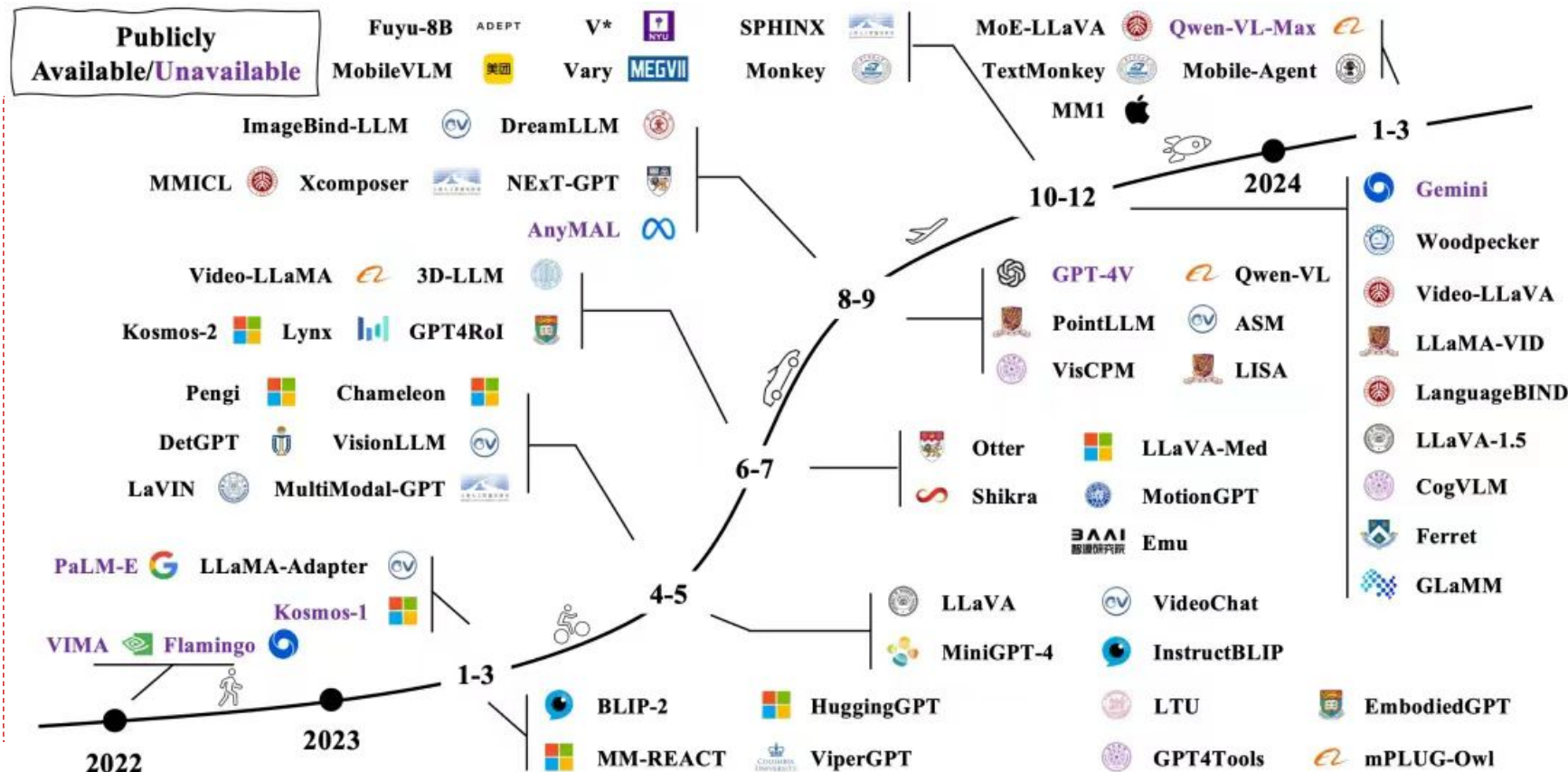


TRANSFORMER"Attention Allows the model to dynamically weigh the importance of different input parts relative to each other.

Understanding Context
Ideal for Machine Translation, Text Generation (ChatGPT), and any task requiring deep contextual understanding.

ChatGPT has pushed AI into the era of Artificial General Intel.

- In November 2022, OpenAI released ChatGPT, which demonstrated unprecedented conversational ability and saw explosive user growth. This ignited a global "large model boom. "
- In China, a "hundred-model war" ensued, with major players like Baidu, Alibaba, Tencent, and Huawei developing their own LLMs, accelerating innovation and expanding AI applications.



The Problem with Early Models: Lack of "Reasoning"



"Parroting" Instead of "Thinking"

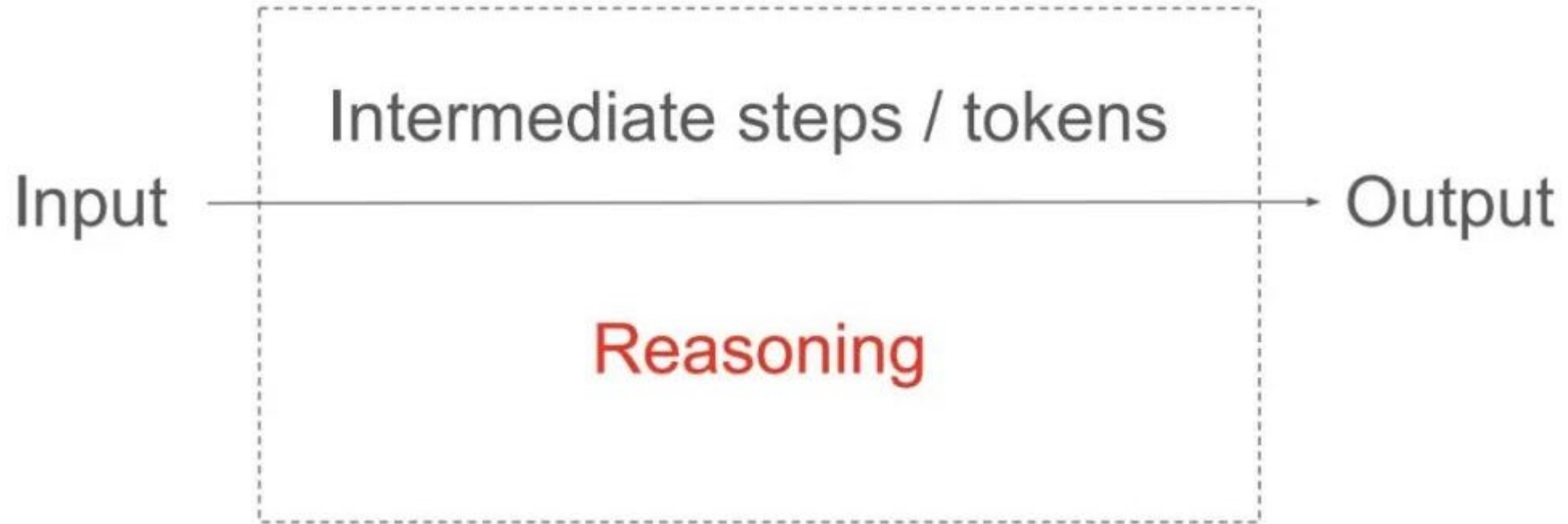
- Early models excelled at regurgitating information and stitching text together, mimicking human language without true understanding.
- They failed at executing step-by-step logic and were highly prone to "hallucinations"—inventing plausible but false facts.
- Critical limitation: They lacked the ability to reason, making them unreliable for complex problem-solving.



The Rise of Reasoning Models (2024)

- Cutting-edge models like OpenAI o1, DeepSeek-R1, and Kimi-k1.5 are engineered to "think step-by-step," decomposing problems logically.
- Capabilities expanded to solve complex multi-step math problems, write high-quality code, and answer advanced scientific questions.
- In rigorous technical evaluations, their proficiency now rivals that of a PhD student in specialized STEM fields.

The inference of LLM



The inference of LLM

No reasoning? Check more generation candidates!

I have 3 apples. My dad has 2 more apples than me. How many apples do we have in total?

5 apples. (Greedy Decoding)

I have 3 apples, my dad has 2 more apples than me, so he has 5 apples. $3+5=8$.

We have 8 apples in total.

You have 3 apples, your dad has 2 more apples than you, so he has 5 apples. $3+5=8$.

The answer is 5.

Xuezhi Wang and Denny Zhou. [Chain-of-Thought Reasoning Without Prompting](#). NeurIPS 2024.

The inference of LLM

Chain-of-Thought Prompting

Q: Elsa has 3 apples. Anna has 2 more apples than Elsa. How many apples do they have together?

A: Anna has 2 more apples than Elsa. So Anna has $2 + 3 = 5$ apples. So Elsa and Anna have $3 + 5 = 8$ apples together.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. [Chain-of-thought prompting elicits reasoning in large language models](#). NeurIPS 2022

Let's Think Step by Step

The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Let's think step by step.

Kojima, T., Gu, S.S., Reid, M., Matsuo, Y. and Iwasawa, Y. Large language models are zero-shot reasoners. NeurIPS 2022.

The inference of LLM

Supervised Finetuning (SFT)

What is the output when concatenating the last letter of each word in “artificial intelligence”? The last letter of “artificial” is “l”. The last letter of “intelligence” is “e”. Concatenating “l” and “e” leads to “le”. So the answer is “le”.

Elsa has 3 apples. Anna has 2 more apples than Elsa. How many apples do they have together? Anna has 2 more apples than Elsa. So Anna has $2 + 3 = 5$ apples. So Elsa and Anna have $3 + 5 = 8$ apples together.

Training data

Finetuning

LLM

Test problem

How many “r”s in “strawberry”?

DeepSeek's Insight: Reasoning Through RL

DeepSeek showed that complex fine-tuning might not be necessary. Instead, **pure reinforcement learning (RL)** could be the key to teaching reasoning.

How it works: The model (agent) attempts to solve problems and receives immediate feedback (reward). Through trial and error, it learns to reason effectively, experiencing "Aha moments."

Benefits: This method is more resource-efficient than traditional fine-tuning and develops robust, generalizable reasoning skills that apply across different tasks.

Open-Source Commitment: DeepSeek released powerful reasoning models that can run on smaller consumer-grade computers, democratizing access to advanced AI capabilities.



Figure: The Reinforcement Learning (RL) Cycle
Agent • Environment • Action • Reward

Volume 645 Issue 8081, 18 September 2025



Subscribe

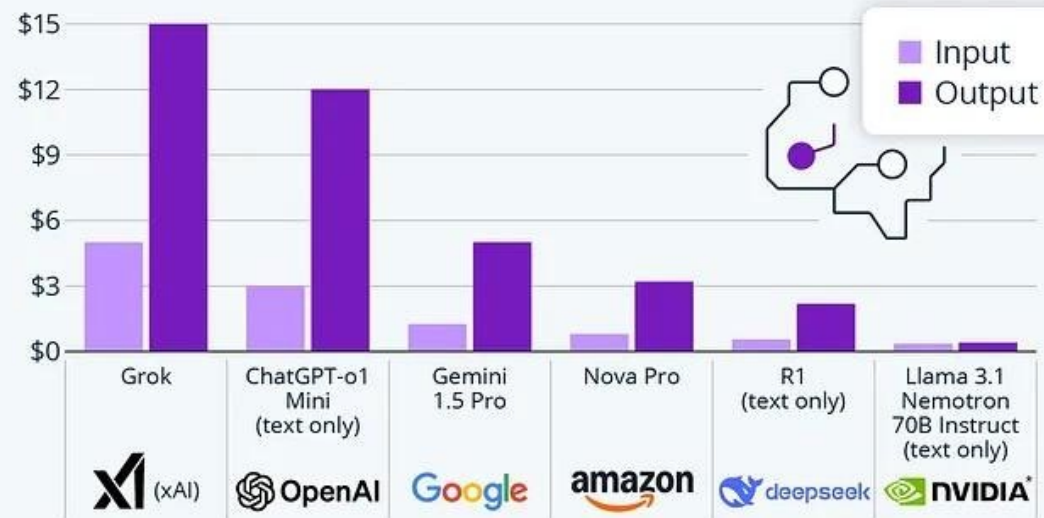
Self-help

Large language models (LLMs) tend to set out the steps they take as they approach more complex challenges for artificial intelligence, similar to how humans approach more complex challenges for artificial intelligence, in the form of annotations. In this week's issue, researchers show a model not only to reason in this way but also to improve itself. The model DeepSeek-R1 was trained using

Cover image: Nik Spencer/ Nature.

DeepSeek-R1 Upsets AI Market With Low Prices

Estimated price for processing one million input/output tokens on different AI models



A token is the smallest unit of AI model processing (~4 characters).
o1 is ChatGPT's latest model. List includes most comparable model per company
* Uses Meta's open-source Llama AI

Source: DocsBot

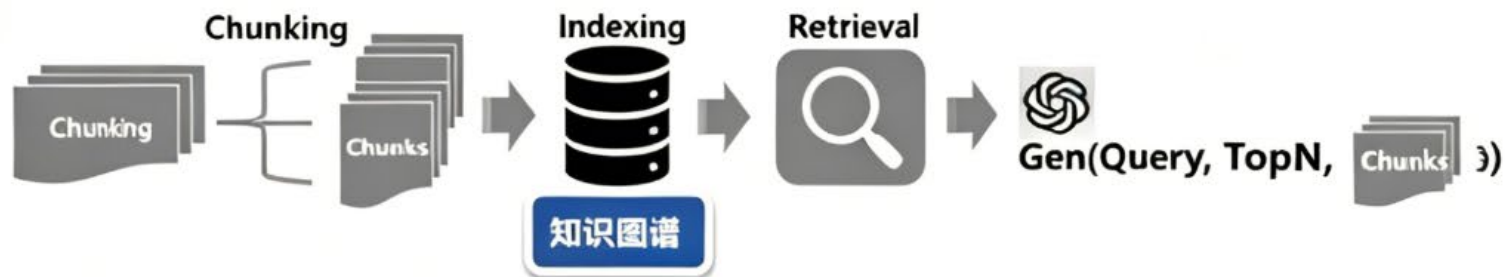


statista

Auxiliary Enhancement: Knowledge Graph+RAG→Private KB

Main Routes of Knowledge-Enhanced Large Models

Evloving based on document retrieval, Docs => R + G **(RAG)**

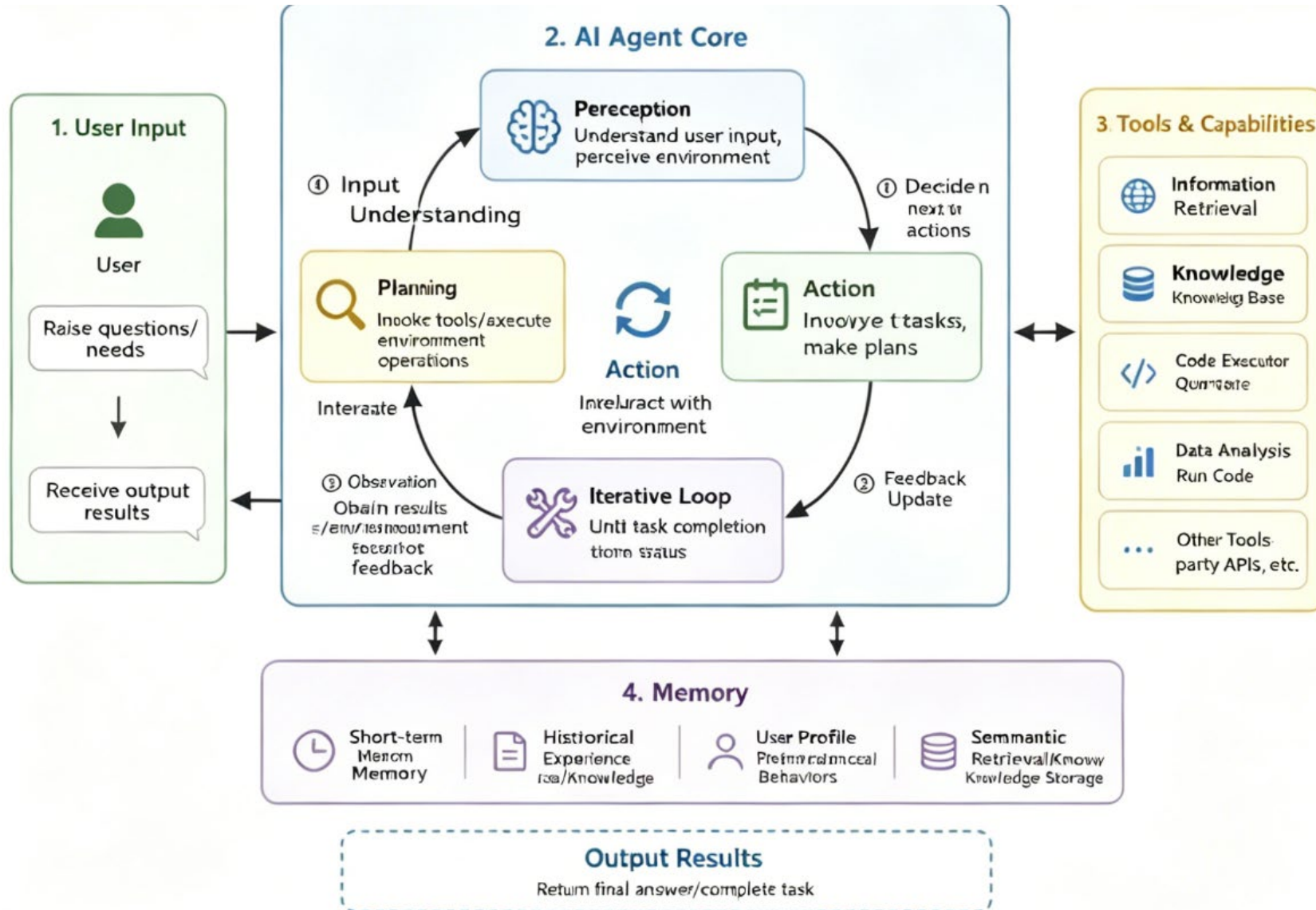


Starting with retrieval (Docs Retrieval) and ending with LLM generation (Generation)

Manus: general AI assistant



Overview of Intelligent Agents



An intelligent agent is a system that perceives the environment, selects actions, and moves towards goals autonomously!